

WPEPlatform API for Android

Web Engines Hackfest

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About me 🙌

- alexg in slack
- long time WebKit contributor working in Linux ports
- **WebKitGTK** and **WPE**
- adapting WebKit to small embedded devices
- currently helping with WPE for Android



Last year at WEH 2025

Adrián's talk: *"Jumping Over the Garden Wall - WPE WebKit on Android"*

- State of **wpe-android** in mid-2025:
 - **Built on upstream** - no downstream patches
 - WPE WebKit **2.48** stable release support · **Android 15**
 - **Three parts**: cerbero → WPE backend → **WPEView** widget



Last year at WEH 2025 - the plan

*Move from the **classic libwpe C API** (verbose, complex) to **WPEPlatform** (default upstream, clear lifecycles, no unnecessary buffer copies, lower-level display sync)*

In this talk we explain the work done to make this transition.



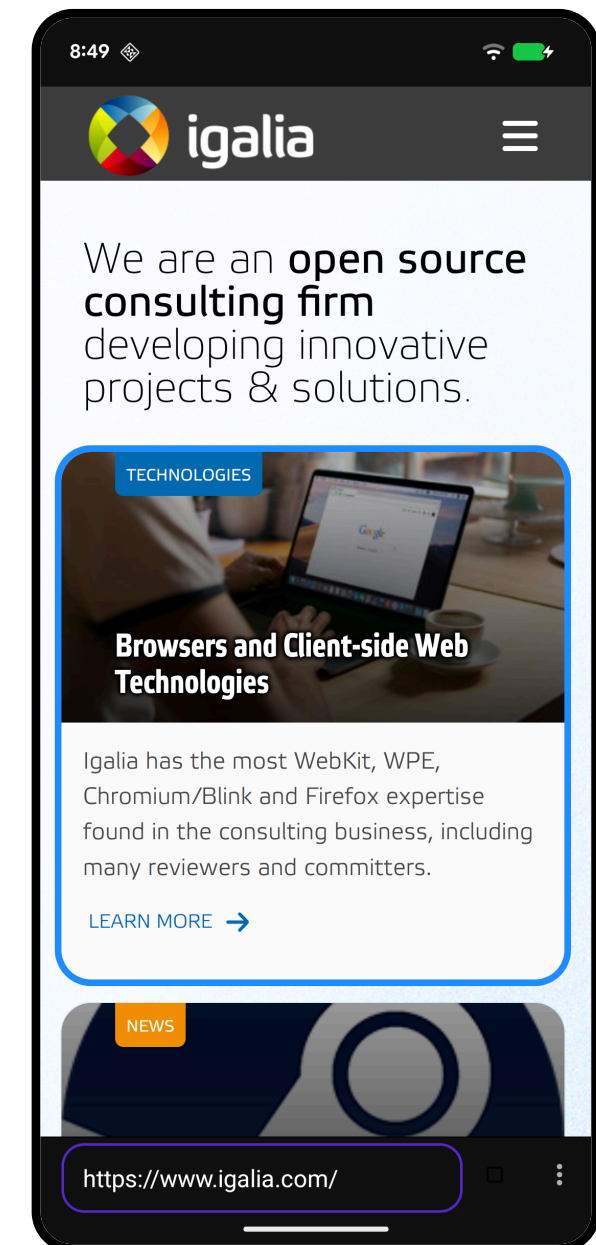
Architecture - libwpe

- Three parts as of mid-2025:
 - **WPEView** Java widget - widget *and* engine handle in one class
 - **WPE Android backend** - platform glue for libwpe backend
 - **Engine** - WebKit + libwpe



Architecture - WPEPlatform

- Five layers (app → engine):
 - **Layer 1 - Convenience API** - `WebView` / `WebContext` (*new*)
 - **Layer 2 - JNI/CAPI bridge** - Proxies for `WebKit` + `WPE`
 - **Layer 3 - WPEPlatform Android backend** (*new*)
 - **Layer 4 - Engine** - upstream `WebKit` + `WPE` port
 - **Layer 0 - Runtime** - `JNI` bootstrap, Android looper, event bridge



The plan

- **Foundation** tooling, cerbero, WebKit bumps
- **Implementation core** WPEPlatform Android backend → CAPI → Convenience API
- **Porting** MiniBrowser, MediaPlayer, WebDriver, Tests
- **Features unlocked** IME, JS dialogs, WebXR
- **Remove the legacy API**



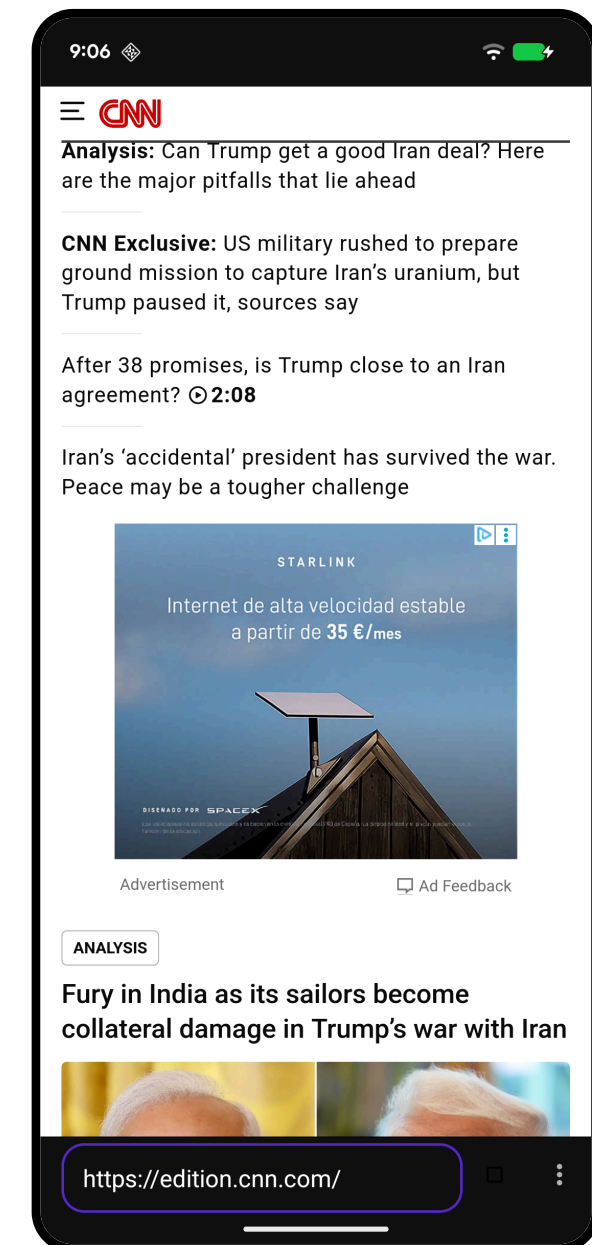
Preparing the ground

- **Dev tooling fixes** - cerbero env in WebKit sdk, NDK install script, bootstrap update documentation
- **Enable WPE Platform API compilation** for Android, still no backend
 - Still some downstream fixes required to make it work (eglGetDisplay issues)
- WebKit bumps now use **main** branch: 2.50.4 → 2.51.93



The WPEPlatform Android backend

- Implement the WPEPlatform abstraction with the NDK:
 - **WPEDisplayAndroid** - EGL init, buffer formats, discovery
 - **WPEToplevelAndroid** - ANativeWindow lifecycle
 - **WPEViewAndroid** - render loop, sync fences, AHardwarebuffer
 - **WPEScreenAndroid** - scale, refresh rate
 - **WPEInputMethodContextAndroid** - IME / text input
 - **WPEScreenSyncObserverAndroid** - vsync via AChoreographer



C API and convenience API

- **C API**: JNI bridge between Java proxy classes and WebKit/WPE GObject
- **Convenience API**: new high-level Android-style Java API
- MiniBrowser: ported first to the new convenience API
- Updated documentation with the architecture



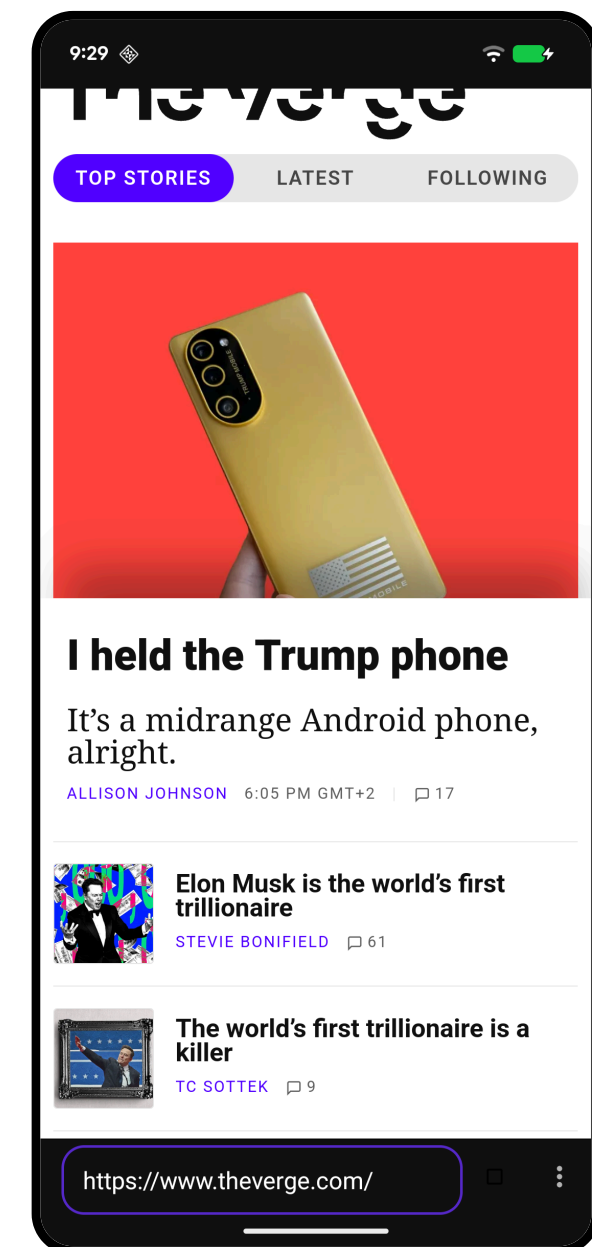
Porting the applications in the repository

- Four applications ported to the new APIs:
 - **MiniBrowser** - our development driver
 - **MediaPlayer** - small media playback example
 - **WebDriver** - browser automation
 - **Tests** - proves API parity
- **Result:** everything in the repository runs on the new API



New features to the API unlocked

- **IME / keyboard** - clean text-input path
- **javascript dialogs** - embedder overrides
- **WebXR / VR** - OpenXR on device
- **Skia compositor** - new default renderer
 - **Deferred Display Lists (DDL)** - GPU-thread replay
- **Scrolling fixes** - smoother touch scrolling



Removing the legacy API

- **Remove the legacy WPEView/WK- API**
 - Safe because every application in the repository uses the convenience API
 - A big amount of code removed - a whole compatibility code is gone
 - Still missing remove libwpe dependency - the Android process launcher code is still there
- **While we were in there:**
 - Fixed UIProcess main thread no longer busy-spinning at 100% CPU
 - Bumped to WPE WebKit new main versions weekly 2.53.3.1



What's next

- **Remove libwpe dependency** - only the process launcher left
- **Permissions API** - geolocation, camera, microphone
- **Screenshot API** - capture the WebView
- **Preedit / composition text** - complete IME support
- **Media & capture state callbacks** - audio, camera, mic, display
- **SSL error callback flow** - per-host accept/reject
- **Improved scrolling** - native touch feel
- **Make Selenium testing CI work** - hardening the development
- **Add Android post-commit bot** - hardening the integration



Numbers + thanks

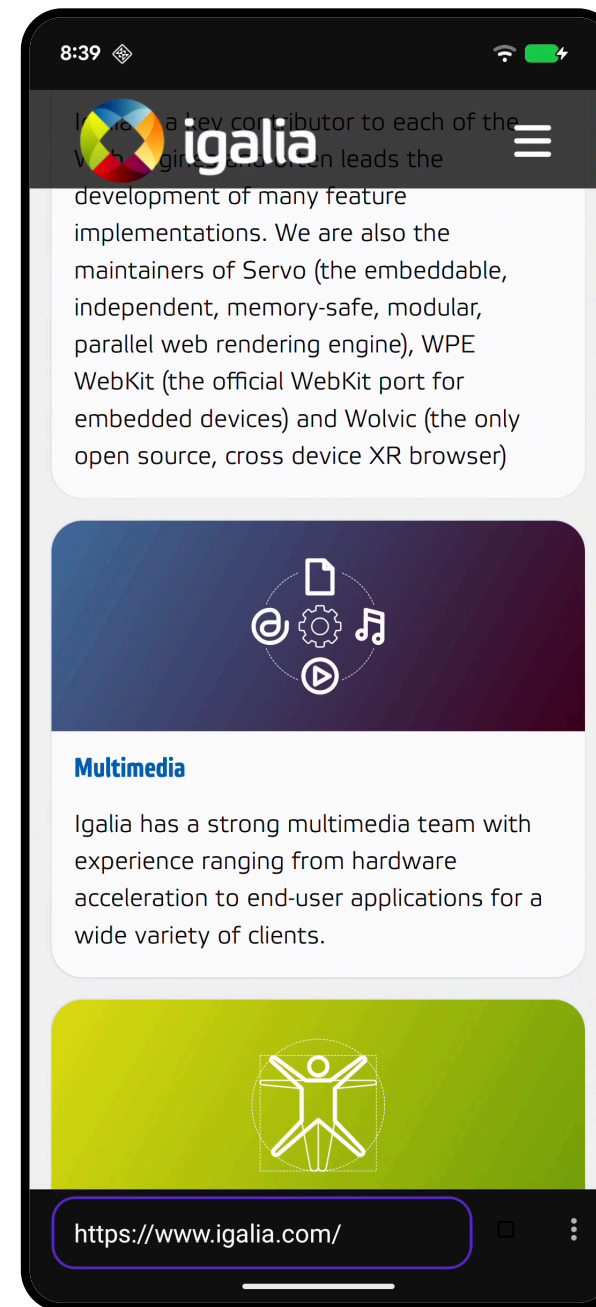
- **45** merged PRs in `wpe-android` since end of 2025
- **28** merged PRs in `wpe-android-cerbero`
- **4** applications ported · WebKit **2.50.4** → **2.53.3.1** · legacy API **deleted**

Thanks: All Igalia teams working around WebKit 🙏

`github.com/Igalia/wpe-android` · `github.com/Igalia/wpe-android-cerbero`



Questions



github.com/Igalia/wpe-android



